

ASSIGNMENT

Numerical Simulation of Excess Carrier Dynamics in Semiconductors

Instructor: ASIF

Using MATLAB / Python

Tool: MATLAB / Python **Submission:** PDF Report + Code (.m / .py) **Total Marks:** 20

Problem 1 — Thermal Relaxation (No Generation, No Bias)

Determine the time behavior of excess carrier concentration as a semiconductor returns to thermal equilibrium.

Conditions:

- n-type semiconductor; $g' = 0$; Zero Bias
- Uniform spatial distribution:

Simulation Tasks:

- Numerically integrate the ODE using Euler or RK4 method
- Plot $\delta p(t)$ vs time t for at least $5 \cdot \tau_p$.
- Overlay the analytical solution and compute the error
- Investigate the effect of varying τ_p on the decay rate

Problem 2 — Transient to Steady State (With Constant Generation)

Determine the time dependence of excess carrier concentration as it reaches steady state under constant illumination.

Conditions:

- p-type; Zero Bias; Uniform spatial distribution

Simulation Tasks:

- Numerically solve the ODE from $t = 0$ using forward Euler or `scipy.integrate.odeint`
- Plot $dp(t)$ vs t showing rise to steady state
- Mark the steady-state value on the plot with a dashed horizontal line
- Show how increasing g_0 affects the steady-state level (parametric sweep)

Problem 3 — Light Pulse: ON/OFF Transient

A semiconductor initially has no excess carriers. Suddenly light shines on its surface, generating excess carriers. After some time the light goes off. Determine the time dependence of excess carrier concentration during both the light-ON and light-OFF phases.

Conditions:

- p-type; Zero Bias; Uniform distribution
- Phase 1 ($0 \leq t \leq t_1$): Light ON — $g' = g_0$,
- Phase 2 ($t > t_1$): Light OFF — $g' = 0$, $dp(t_1) = \text{initial condition}$

Simulation Tasks:

- Simulate both phases in sequence using a single numerical solver
- Plot $dp(t)$ across both phases (use different colors or shading per phase)
- Mark the transition time t_1 clearly on the plot
- Investigate effect of t_1 duration on the peak excess carrier value

Problem 4 — Steady-State Spatial Distribution

Determine the steady-state spatial distribution of excess carrier concentrations under surface illumination at $x = 0$.

Conditions:

- n-type; Zero Bias; Steady state ($d/dt = 0$)

Simulation Tasks:

- Numerically solve the BVP using finite difference or `scipy.integrate.solve_bvp`. Plot $dp(x)$ vs x showing exponential decay into the bulk.
- Overlay the analytical solution on the same axes
- Show the effect of varying L_p (diffusion length) on the spatial extent

Problem 5 — Full PDE: Time + Space Dependence

Determine both the time dependence and spatial dependence of excess carrier concentrations simultaneously.

Conditions:

- n-type; Transient with diffusion, recombination, and generation; $g=0$;

Simulation Tasks:

- Implement finite-difference in x (Method of Lines), solve in time with `odeint` / `ode45`
- Generate a 2D surface/color plot of $dp(x, t)$
- Plot dp vs x at several fixed time snapshots ($t = 0.5\tau_p, \tau_p, 2\tau_p, 5\tau_p$)
- Plot dp vs t at several fixed spatial positions ($x = 0, L_p, 2L_p$)
- Show convergence to the steady-state solution from Problem 4 as $t \rightarrow \infty$

Suggested Parameters

Parameter	Symbol	Suggested Value	Unit
Minority carrier lifetime	τ_p	1e-6	s
Diffusion coefficient	D_p	10	cm ² /s
Diffusion length	$L_p = \sqrt{D_p \tau_p}$	1e-2	cm
Surface excess carrier	dp_s	1e15	cm ⁻³
Generation rate	g_0	1e21	cm ⁻³ s ⁻¹
Spatial domain	x	0 to $5L_p$	cm
Time domain	t	0 to $5\tau_p$	s

Marking Rubric

Criterion	Applies To	Marks
Correct physics and governing equations	All problems	5
Accurate numerical implementation	Problems 1-5	7
Plots: labeled axes, title, legend	All problems	5
Code clarity and comments	All problems	2
Brief written explanation in report	All problems	1

Submission Instructions

- One MATLAB (.m) or Python (.py) script per problem, clearly commented
- PDF report with all plots labeled (axes, title, legend) and 2-4 sentence explanation each
- Submit a single ZIP file named: StudentID_Assignment2.zip